Nicolas Kass

Software, data & project consultant

I build the operational systems small businesses actually run on — inventory, point of sale, invoicing, field tooling — and the data pipelines behind them. Backend, frontend, deployment and the operation itself, because for most of my career there was nobody else to hand it to. I came to software from field biology, which is where I learned that a number needs a defence.

La Plata, Argentina · remote-friendly   github.com/nicolaskass   LinkedIn   t3.com.ar

WHAT I HAVE PUBLISHED

Four public repositories documenting how my systems are built and why. No client source code — the reasoning: what I decided, what I rejected, and what each decision cost. **34 architecture decision records** between them.

Specialty Retail ERP

github.com/nicolaskass/specialty-retail-erp

A production ERP · 13 modules · 84 models · 229 routes · ~190k LOC

System architecture at real scale: modules as bounded contexts with an acyclic dependency graph, and a consolidation refactor that unified four transaction types and three payment systems over live financial data.

Django · DRF · MongoDB · React · Docker · Traefik

Geospatial Survey Toolkit

github.com/nicolaskass/geospatial-survey-toolkit

Field tooling for a land-surveying practice

Road-works quality control, survey-code validation and a raster-to-CAD vectoriser, delivered to surveyors on Windows laptops with no Python and sometimes no internet. The tool flags anomalies and refuses to correct them: field data is evidence.

Python · Shapely · Streamlit · ezdxf · potrace

Geospatial Data Processing

github.com/nicolaskass/geospatial-data-processing

LiDAR & photogrammetry pipeline · reference book, 67 chapters

Measurement discipline applied to terrain data: validation against check points never used as control, uncertainty propagated to the delivered volume as two honest bounds, and quality gates that refuse rather than return a plausible wrong answer.

LiDAR · Photogrammetry · PDAL · GDAL · rasterio · Quarto

Ecological Data Analysis

github.com/nicolaskass/ecological-data-analysis

Statistics behind four manuscripts · 4 published reproducibility packages

Bayesian capture–recapture implemented directly and cross-checked against an established reference, with prior sensitivity analysis and a simulation of my own estimator's bias. Every reported number is generated, never transcribed.

Bayesian inference · MCMC · NumPy · SciPy · Reproducibility

HOW I WORK

The parts that are not standard. Django and React do not distinguish anyone; these do.

I make drift impossible instead of promising to prevent it

A post-commit hook that flags documentation gone stale against the code. A checker that verifies every citation from code to manuscript still resolves. A generated file that a paper reads its numbers from, so text and tables cannot disagree. Three domains, three times the same conclusion: anything maintained by intention decays — bind it mechanically.

I meter what I spend

Every LLM call in production writes tokens, latency and estimated cost to a usage log priced from a maintained table. Cost per feature is a query, not a surprise at the end of the month. Inference is a purchase, and systems meter what they buy.

I design what happens when a dependency is gone

Optional integrations degrade instead of failing: the provider SDK is imported at the call site, every entry point returns a structured result rather than raising, and the business flow completes with one field empty. A third-party integration has sat broken in production for months, by design, with no user-visible failure.

I know where automation should stop

In the surveying tools the software classifies anomalies across three severity levels and corrects none of them, because a miscoded point is sometimes a typo and sometimes the only real defect in the job. Automating a judgement is only safe when being wrong is cheap.

I write down what a decision cost

Every decision record has a section for its downsides, including the cases where I would choose differently for a different problem. A document that lists only benefits is marketing, and it makes everything around it less believable.

EXPERIENCE

Founder · Process & systems consultant

T³ — Three Tech Thinkers · La Plata, Argentina

2023 – present

- Operational diagnosis for small businesses: process mapping, quantification of time and money lost, and an intervention plan prioritised by impact.
- End-to-end project direction: scope, implementation sequence, stakeholder management, and evaluation against the objectives set in the diagnosis.
- Full-stack development of custom management systems — Python/Django and Node.js on the backend, React with Vite on the frontend, MongoDB and SQL, REST APIs and third-party integrations.
- LLM APIs integrated into production systems, with per-call cost metering and graceful degradation so an outage never blocks the business flow.
- Own infrastructure: Linux, Docker and Compose, Traefik and Nginx, VPS deployment with SSL, CI/CD on GitHub Actions.
- Process redesign and documentation under ISO 9001 criteria.

Head of operations & systems development

Digital Discos · La Plata, Argentina

2022 – 2026 · full responsibility from 2024

- Directed the complete operation of a 30-year-old family retail business: purchasing, sales, customer service, staff, stock and suppliers.
- Designed and built the business's management system: inventory built for 200,000+ *unique* catalogue items — distinct records with their own price and stock, not repeated units.
- Marketplace API integration for listing, price control and monitoring against the real market.
- Automated the order cycle end to end: intake with live catalogue, price and availability, and sale processing without manual intervention.
- The business went from selling only over the counter to operating nationally without growing its structure.

University lecturer & researcher

Universidad Nacional de La Plata (UNLP) · Argentina

2008 – present · formal appointment since 2010

- Research in natural sciences and undergraduate teaching. Design of studies with no prior art, data analysis, and communication of results.
- Entered through research scholarships, internships and teaching-assistant collaboration in 2008; formally appointed in both research and teaching from 2010.
- PhD candidate in Natural Sciences. Four manuscripts in review or preparation, each with a public reproducibility package.
- This is the direct origin of the method I apply in consulting: understand the system before intervening, measure against a defined objective, and adjust on evidence rather than intuition.

External process consultant · ISO 9001

Independent

2022

- Certified Internal Auditor, ISO 9001:2015 Quality Management Systems — Bureau Veritas Argentina, May 2022.
- Redesign of operational processes and procedures under ISO 9001, identification and quantification of unrecorded time and financial losses, and quality management system documentation.

Field technician · Antarctic ecosystem monitoring

Instituto Antártico Argentino — Dirección Nacional del Antártico · CCAMLR Ecosystem Monitoring Programme (CEMP)

September 2018 – March 2019 · Antarctic summer pre-campaign and summer campaign

- Ecosystem monitoring under the CEMP protocols of CCAMLR, the international commission that manages Antarctic marine living resources. The data collected feeds Argentina's reporting to that commission, where it informs fisheries management decisions.
- Standardised sampling procedures, instrument calibration, and documented data custody, to a protocol written to be comparable across countries and across decades — the whole point of the programme is that a measurement taken in one season can be compared to one taken thirty years earlier.
- Antarctic field seasons are unforgiving of improvisation: a sample cannot be retaken and the season does not come back. This is where protocol discipline stopped being an abstraction for me, and it is the direct ancestor of how I treat measurement, validation and evidence in software.

Alumni · Project co-leader

Conservation Leadership Programme

2020 · 2022

- International training programme for conservation leaders. Tools from the course: project planning and management, logical framework with problem-tree analysis, stakeholder analysis and engagement, and strategic communication. Global virtual cohort in 2020; in-person in South Africa in 2022, with LATAM and Africa cohorts.
- Future Conservationist Award as team co-leader.

TECHNICAL SKILLS

Backend	Python · Django · Django REST Framework · Node.js · Express · REST API design · JWT and OAuth · MongoDB / MongoEngine · SQL
Frontend	React · Vite · TypeScript · JavaScript · Tailwind · TanStack Query · HTML/CSS
Infrastructure	Linux · Docker & Compose · Traefik v3 · Nginx · VPS operation · Let's Encrypt · GitHub Actions · MCP servers
Geospatial	GDAL/OGR · PDAL · rasterio · Shapely · GeoPandas · LiDAR classification · photogrammetry (SfM/MVS) · DTM/DSM · DXF · GNSS RTK/PPK
Data & statistics	NumPy · SciPy · pandas · Bayesian inference and MCMC · capture–recapture models · non-parametric methods · uncertainty propagation · Matplotlib
Practice	Architecture decision records · modular monolith design · schema migration discipline · testing (pytest, Vitest, Playwright) · documentation as code (MkDocs, Quarto) · ISO 9001
AI tooling	Anthropic API in production · agent orchestration with version-controlled role definitions and per-role model-cost tiers · usage metering and cost attribution

PROJECT MANAGEMENT & LEADERSHIP

Projects	Project planning and management · logical framework and problem-tree analysis · stakeholder analysis · stakeholder engagement · operational diagnosis with impact-based prioritisation · process design under ISO 9001
Leadership	Team co-leader in the Conservation Leadership Programme (Future Conservationist Award) · ran a retail business end to end: staff, suppliers, purchasing and sales · client projects led from diagnosis to delivery
Communication	University teaching since 2010 · strategic communication (CLP) · communicating scientific results · architecture decision records and technical documentation

EDUCATION & LANGUAGES

PhD candidate, Natural Sciences

Universidad Nacional de La Plata · conservation of Patagonian amphibians

In progress

Licenciado en Biología (Zoology) — equivalent to MSc

Universidad Nacional de La Plata

2016

Certified Internal Auditor — ISO 9001:2015

Bureau Veritas Argentina · 2022

Languages

Spanish — native · English — professional working proficiency · German — B1 (certified)

Academic record

Publications, grants, teaching and conferences: ResearchGate. Full academic CV available on request.

Software

Programming since childhood and taught throughout school; self-taught since, working on Linux exclusively since around 2010. No formal computing qualification — the repositories above are the argument.

Every figure quoted here is a real measurement taken from the projects — lines, modules, routes and records counted from the source, not estimated. Client identifiers and deployment hostnames are omitted throughout.

Last updated: September 2026 · Versión en español